

Machine Learning for Natural Language Processing

Training Language Models with Reinforcement Learning



UNIVERSITY OF
GOTHENBURG

CHALMERS

Richard Johansson

`richajo@chalmers.se`

Training language models to follow instructions with human feedback

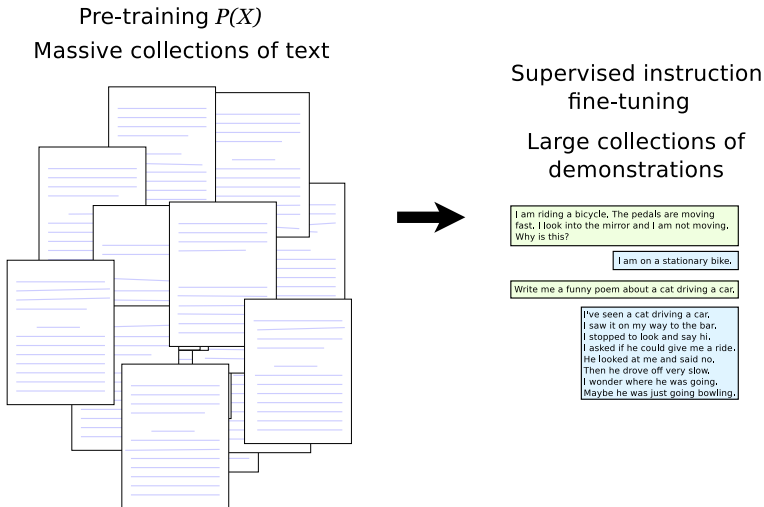
Long Ouyang* Jeff Wu* Xu Jiang* Diogo Almeida* Carroll L. Wainwright*
Pamela Mishkin* Chong Zhang Sandhini Agarwal Katarina Slama Alex Ray
John Schulman Jacob Hilton Fraser Kelton Luke Miller Maddie Simens
Amanda Askell† Peter Welinder Paul Christiano*†
Jan Leike* Ryan Lowe*

OpenAI

Abstract

Making language models bigger does not inherently make them better at following a user's intent. For example, large language models can generate outputs that are untruthful, toxic, or simply not helpful to the user. In other words, these models are not *aligned* with their users. In this paper, we show an avenue for aligning language models with user intent on a wide range of tasks by fine-tuning with human feedback. Starting with a set of labeler-written prompts and prompts submitted through the OpenAI API, we collect a dataset of labeler demonstrations

Starting point: Pre-training and supervised instruction tuning



Problem: How to train for high-level quality

What is the name of the capital of Sweden?

Stockholm

Write a Python one-liner that finds the longest string in a list of strings.

```
max(the_list, key=len)
```

```
max((len(s), s) for s in the_list)[1]
```

⋮

Write a slogan about the values of Chalmers.

The core value of Chalmers is free expression.

We stand for inclusiveness, creativity, and sustainability.

⋮

Reinforcement learning: A more suitable framework?

Maximum likelihood of individual tokens may be too primitive

On a high level, RL optimizes the model to maximize the
expected **reward**

$$\max_{\phi} \mathbb{E}[\mathbf{R}]$$

The reward can implement our notion of “text quality”

Reinforcement learning: A more suitable framework?

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The reward can implement our notion of “text quality”

But: how do we define the reward?

Idea: learn a **reward model** from **human preferences**

Generating and ranking

Write a slogan about the values of Chalmers.

The values of Chalmers are limitless.

I don't understand your request.

We stand for inclusiveness, creativity, and sustainability.

Generating and ranking

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We stand for inclusiveness, creativity, and sustainability.

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The Bradley–Terry binary preference model

$$P(y_1 \text{ preferred over } y_0) = \sigma(\mathbf{r}_\theta(x, y_1) - \mathbf{r}_\theta(x, y_0))$$

InstructGPT initialized the reward model from a pre-trained GPT

Reinforcement learning and policy gradient methods

There is a great variety of methods in RL building on many theoretical frameworks

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Policy gradient methods compute (approximate) gradients in the direction of increasing the reward

Intuition: sample some sequences and adjust the model to prefer high-reward sequences

Proximal Policy Optimization

Many RL training methods suffer from stability problems

InstructGPT uses a policy gradient method called **Proximal Policy Optimization** (PPO) that includes several tricks for stabilization

PPO in InstructGPT

$$\begin{aligned} \text{obj}(\phi) = & \mathbb{E}_{(x,y) \sim D_{\pi_{\phi}^{\text{RL}}}} [\mathbf{r}_{\theta}(x,y) - \beta \log(\pi_{\phi}^{\text{RL}}(y|x) / \pi^{\text{SFT}}(y|x))] \\ & + \gamma \mathbb{E}_{x \sim D_{\text{pretrain}}} [\log \pi_{\phi}^{\text{RL}}(x)] \end{aligned}$$

For the implementation details...

Secrets of RLHF in Large Language Models Part I: PPO

Rui Zheng^{*†}, Shihan Dou^{*†}, Songyang Gao^{*},

Yuan Hua[‡], Wei Shen, Binghai Wang, Yan Liu, Senjie Jin, Qin Liu,
Yuhao Zhou, Limao Xiong, Lu Chen, Zhiheng Xi, Nuo Xu, Wenbin Lai,
Minghao Zhu[‡], Cheng Chang, Zhangyue Yin, Rongxiang Weng,
Wensen Cheng, Haoran Huang[‡], Tianxiang Sun, Hang Yan,

Tao Gui[‡], Qi Zhang[‡], Xipeng Qiu, Xuanjing Huang

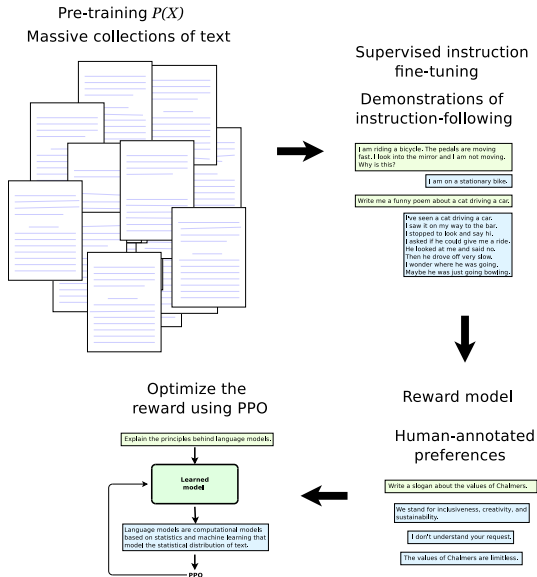
Fudan NLP Group

[‡] ByteDance Inc

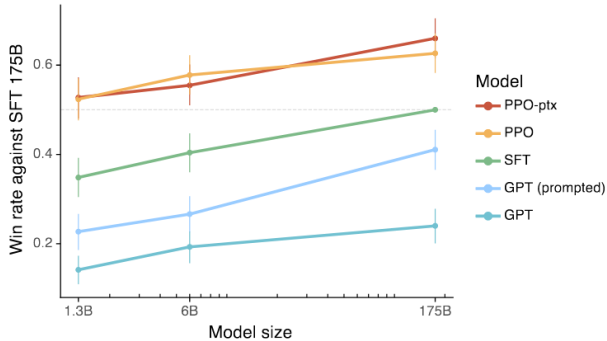
Abstract

Large language models (LLMs) have formulated a blueprint for the advancement of artificial general intelligence. Its primary objective is to function as a human-centric (helpful, honest, and harmless) assistant. Alignment with humans assumes paramount significance, and reinforcement learning with human feedback (RLHF) emerges as the pivotal technological paradigm underpinning this pursuit. Current technical routes usually include **reward models** to measure human preferences, **Proximal Policy Optimization (PPO)** to optimize policy model outputs, and **process supervision** to improve step-by-step reasoning capabilities. However, due to the challenges of reward design, environment interaction, and agent training, coupled with huge trial and error cost of large language models, there is a significant barrier for AI researchers to motivate the development of technical alignment and safe landing of LLMs. The stable training of RLHF has still been a puzzle.

Full LLM training procedure: Summary



What is the effect of RLHF?



More reading and open-source RLHF implementations

HuggingFace's tutorial gives a description of RLHF and some open-source implementations

The **TRL** library allows RLHF to be applied to HuggingFace models

<https://github.com/huggingface/trl>

references

- L. Ouyang, J. Wu, and X. Jiang et al. 2022. [Training language models to follow instructions with human feedback](#). arXiv:2203.02155.
- R. Zheng, S. Dou, and S. Gao et al. 2023. [Secrets of RLHF in large language models. Part I: PPO](#). arXiv:2307.04964.